THE VALUATIVE CRITERION FOR SEPARATEDNESS OF MORPHISMS OF ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement carries an optional name argument ('Valuative criterion separatedness'); the optional argument is part of the counted statement span and is preserved verbatim.

Editorial note. The four references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Quasi-separatedness of f is quasi-compactness of the diagonal. used at proof lines 9217-9218. Supplies one of the two inputs of the converse criterion.
- The converse criterion: quasi-compact plus existence part implies universally closed. Lemma lemma-quasi-compact-existence-universally-closed, used at proof lines 9232-9234. Makes the diagonal universally closed once the criterion is verified for it.
- A universally closed, locally finite type, separated representable monomorphism is a finite morphism. More on Morphisms, Lemma more-morphisms-lemma-characterize-finite, with Spaces, Lemma spaces-lemma-representable-transformations-property-implication, used at proof lines 9235-9241. Upgrades universal closedness to finiteness of the diagonal.
- A finite monomorphism is a closed immersion. Morphisms, Lemma morphisms-lemma-finite-monomorphism-closed, used at proof lines 9242-9244. Concludes that the diagonal is a closed immersion, i.e. f is separated.

Source reference. Original lemma 03KA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4670–4670. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02LS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3419–3419. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03BB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4342–4342. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 02YO: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3883–3883. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2 (Valuative criterion separatedness)**.** *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Assume*

(1) *the morphism f is quasi-separated, and*
(2) *the morphism f satisfies the uniqueness part of the valuative criterion.*

Then f is separated.

Proof. Assumption (1) means $\Delta_{X/Y}$ is quasi-compact. We claim the morphism $\Delta_{X/Y} : X \to X \times_Y X$ satisfies the existence part of the valuative criterion. Let a solid commutative diagram

$$\begin{array}{ccc} \mathrm{Spec}(K) & \longrightarrow & X \\ \downarrow & \nearrow & \downarrow \\ \mathrm{Spec}(A) & \longrightarrow & X \times_Y X \end{array}$$

be given. The lower right arrow corresponds to a pair of morphisms $a, b : \mathrm{Spec}(A) \to X$ over Y. By assumption (2) we see that $a = b$. Hence using a as the dotted arrow works. Hence Lemma 03KA applies, and we see that $\Delta_{X/Y}$ is universally closed. Since always $\Delta_{X/Y}$ is locally of finite type and separated, we conclude from More on Morphisms, Lemma 02LS that $\Delta_{X/Y}$ is a finite morphism (also, use the general principle of Spaces, Lemma 02YO). At this point $\Delta_{X/Y}$ is a representable, finite monomorphism, hence a closed immersion by Morphisms, Lemma 03BB. □